

Lecture 22: Point Estimation for Mean and Variance

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Table of contents

- 1 Estimators, bias, variance, MSE::: **Definition: Estimator, bias, MSE** An estimator $\hat{\theta}(X_1, \dots, X_n)$ of θ is a random variable, with

$$\text{Bias}(\hat{\theta}) = E[\hat{\theta}] - \theta, \quad \text{MSE}(\hat{\theta}) = E[(\hat{\theta} - \theta)^2] = \text{Bias}(\hat{\theta})^2 + \text{Var}(\hat{\theta}).$$

Fact $\bar{X}_n$ is an unbiased estimator of μ . The sample variance $S_n^2 = \frac{1}{n-1} \sum_i (X_i - \bar{X}_n)^2$ (note $n-1$!) is unbiased for σ^2 . Bias correction: why $n-1$, not n ? 1

—title: “Lecture 22: Point Estimation for Mean and Variance”author: “Jamie Haddock”format: revealjs: output-file: Lecture22_slides slide-number: true preview-links: auto logo: figs/hmc.png css: input/slides.css incremental: true smaller: true code-fold: true embed-resources: true html: code-fold: true embed-resources: true pdf: documentclass: article toc: true number-sections: true geometry: - top=1in - left=1in - bottom=1in - right=1informat-links: falsejupyter: python3filters: - input/remove-pause.luaexecute: echo: true eval: true—

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```
mu, sigma2, n = 2.0, 5.0, 8biased, unbiased = [], []for _ in range(30000): X = rng.normal(mu, np.sqrt(sigma2))
```

—# Case Study: Randomized Matrix Multiplication as an Estimator $Z = \frac{1}{p_i} a_i b_i^\top$ is unbiased for AB , so $\text{MSE}(Z) = \text{Var}(Z)$. The optimal-sampling distribution from L16 is literally the **minimum-MSE** unbiased estimator in this family.

```
n_, m, p_dim = 6, 50, 5A = rng.normal(size=(n_, m))B = rng.normal(size=(m, p_dim))true_AB = A @ Bdef ms
```

Midterm 2 is Thursday, covering Weeks 7-12.## Reading & HomeworkReading: §9 (introduction) (Bertsekas & Tsitsiklis, Introduction to Probability).Due today: Homework 10 (Gradescope).